ASSEMBLY OF THE LOCAL TYPE II EXCLUSION THEOREM FOR NAVIER–STOKES BLOWUP

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Abstract. We assemble the local Type II exclusion argument for the three-dimensional incompressible Navier–Stokes equations in standard PDE language. The input is a suitable weak solution near a putative singular point, the local Caffarelli–Kohn–Nirenberg concentration theorem, a local Type I tangent criterion, and the five preceding PDE results in the series: the local single-core Type II criterion, the localized scale-translation representation, the local multibubble and cascade exclusion, the compact-cylinder Caccioppoli regularity theorem, and the scale-collapse dichotomy. Under the hypotheses stated in those results, every positive local Type II concentration is exhausted by the single-core, multibubble/cascade, rough-core, or scale-collapse alternatives, and each alternative is excluded by the corresponding theorem. Consequently no Type II singularity occurs at the point under these analytic hypotheses.

1. Introduction

Let (u, p) be a suitable weak solution of the three-dimensional incompressible Navier–Stokes equations

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \nabla \cdot u = 0,$$

in a neighborhood of a possible singular point $z_0 = (x_0, T)$. This paper assembles the local Type II part of the analysis. It adds no new analytic argument; it explains how the local entry result together with the five PDE papers in the series combine into one exclusion theorem.

The entry point is local. For $r > 0$, set

$$Q_r(z_0) = B_r(x_0) \times (T - r^2, T),$$

and define the pressure-normalized Caffarelli–Kohn–Nirenberg quantities

$$C(z_0, r) = r^{-2} \int_{Q_r(z_0)} |u|^3 \, dx \, dt,$$

$$D(z_0, r) = r^{-2} \int_{T-r^2}^{T} \int_{B_r(x_0)} |p(x,t) - (p)_{B_r(x_0)}(t)|^{3/2} \, dx \, dt.$$

The pressure is always understood modulo functions of time, as in the local energy inequality and the CKN theory [CKN82].

The local concentration theorem says that a singular point has positive local CKN concentration. After the local Type I tangent criterion removes nonzero bounded parabolic tangent limits, the remaining positive concentration is the local Type II regime. The preceding papers then divide that regime into the following PDE alternatives.

(i) A single selected core with bounded compact-cylinder suitable package and nondegenerate localized scale-translation gauges.
(ii) A multibubble, cascade, or gauge-degenerate configuration.
(iii) Failure of local windowed H^1 control in a bounded renormalized core.
(iv) Collapse of the renormalized scale.

The single-core alternative is excluded by the local compact criterion. The multibubble and cascade alternatives are excluded by the local decoupling and scale-rigidity theorems. The rough-core alternative is excluded by the compact-cylinder Caccioppoli theorem. The scale-collapse alternative is excluded either by a finite-cost condition controlled by the scale-drift cost or by reduction to a stationary or generalized self-similar profile in a parameter regime covered by the relevant Liouville theorem.

Theorem 1.1 (Local Type II exclusion, assembled form)**.** *Let (u, p) be a suitable weak solution near (x_0, T). Assume the local Type I tangent criterion. Assume also the analytic hypotheses of the five preceding PDE theorems in the series: local single-core exclusion, localized scale-translation representation, local multibubble/cascade exclusion, compact-cylinder Caccioppoli regularity, and scale-collapse exclusion by cost or by stationary-profile rigidity. Then (x_0, T) is not a Type II singular point.*

The theorem is deliberately conditional on the analytic hypotheses named above. It does not assert global regularity for arbitrary initial data. Its conclusion is that, once the local Type I criterion and the five local Type II components are available in their stated forms, the Type II alternatives are exhausted and excluded.

2. Local entry and Type I separation

Theorem 2.1 (Positive local concentration at singular points)**.** *If (x_0, T) is singular, then*

$$\limsup_{r\downarrow 0}\big(C((x_0,T),r) + D((x_0,T),r)\big) > 0.$$

Equivalently, there are $\eta > 0$ and $r_n \downarrow 0$ such that

$$C((x_0,T),r_n) + D((x_0,T),r_n) \geq \eta \qquad \text{for all } n.$$

Proof. This is the contrapositive of CKN epsilon regularity. If the displayed limsup were zero, then for sufficiently small r the CKN quantity would be below the universal regularity threshold, and (x_0, T) would be regular. □

Given such a sequence, define the parabolic rescalings

$$u_n(y,s) = r_n u(x_0 + r_n y, T + r_n^2 s), \qquad p_n(y,s) = r_n^2 p(x_0 + r_n y, T + r_n^2 s).$$

Then (u_n, p_n) is suitable on every fixed backward cylinder once n is large enough, and the unit cylinder retains positive normalized CKN mass.

Assumption 2.2 (Local Type I tangent criterion)**.** Any blowup sequence whose selected-window rescalings have a nonzero bounded parabolic tangent on a compact cylinder is excluded by the Type I theory.

A positive local concentration sequence not excluded by this criterion is called a local Type II sequence in this paper.

3. Compact-cylinder dichotomy

For a suitable pair (v, q) on a backward cylinder, write

$$A_v(R) = R^{-1} \operatorname*{ess\,sup}_{-R^2<s<0} \int_{B_R} |v(y,s)|^2 \, dy,$$

$$E_v(R) = R^{-1} \int_{-R^2}^{0} \int_{B_R} |\nabla v|^2 \, dy \, ds,$$

and let $C_{v,q}(R)$, $D_{v,q}(R)$ denote the corresponding scaled local L^3 and pressure quantities.

Theorem 3.1 (Local compactness dichotomy). *After passing to a subsequence of the local rescalings (u_n, p_n), exactly one of the following alternatives holds.*

(i) *For every fixed $R < \infty$,*

$$\sup_n \left(A_{u_n}(R) + E_{u_n}(R) + C_{u_n,p_n}(R) + D_{u_n,p_n}(R)\right) < \infty.$$

In this case a subsequence converges locally in the suitable weak class on compact backward cylinders.

(ii) *For some fixed $R < \infty$, the preceding supremum is infinite. Then a bounded rescaled cylinder carries an additional local concentration, giving a multibubble or cascade alternative.*

Proof. The first alternative is precisely the compact-cylinder suitable package. If it holds, the standard compactness theorem for suitable weak solutions gives a subsequence with local weak suitable convergence, strong convergence in subcritical local spaces, and pressure convergence modulo spatial means. If it fails, then at least one component of the local suitable package is unbounded on a fixed cylinder. The local Caccioppoli inequality shows that boundedness of $C + D$ on a comparable larger cylinder controls $A + E$ on a smaller one. Thus failure of the compact-cylinder package forces additional bounded-cylinder concentration, which is the local multibubble or cascade alternative. □

4. Imported PDE components

This section lists the five PDE components used in the assembly. They are not reproved here.

Theorem 4.1 (Local single-core criterion). *Assume the local Type I tangent criterion. A local Type II sequence with one selected core, bounded compact-cylinder suitable package, nondegenerate localized scale-translation gauges, compact-ball pressure decomposition, local Caccioppoli bounds, and finite localized Type II cost cannot occur.*

Proof recalled from Paper I. Finite localized cost gives windows on which the localized dissipation tends to zero. Compact-cylinder bounds give convergence on the same windows. The low-dissipation limit is spatially constant on the selected core. If the constant is zero, the selected CKN mass vanishes, contradicting positive local concentration. If the constant is nonzero, the sequence has a nonzero bounded parabolic tangent and is excluded by the local Type I tangent criterion. □

Theorem 4.2 (Localized scale-translation representation). *On a genuine single-core local Type II sequence, the localized scale and centering conditions can be imposed on compact cylinders whenever the localized Jacobian is nondegenerate. In the resulting variables (V, P),*

$$\partial_\tau V + (V\cdot\nabla)V + \nabla P = \nu\Delta V + a(\tau)(V + y\cdot\nabla V) + b(\tau)\cdot\nabla V, \qquad \nabla\cdot V = 0,$$

and the pressure is determined modulo functions of time by the compact-ball pressure decomposition. Failure of the localized nondegeneracy condition is a multibubble or gauge-degenerate alternative.

Proof recalled from Paper II. The localized scale and centering functionals have an invertible derivative on the selected core. The quantitative inverse function theorem gives the local scale-translation choice on each compact good window. Differentiating the rescaled variables gives the displayed equation, and the pressure transforms with the usual freedom to add functions of time. If the derivative is not invertible, a unique core has not been selected, which is precisely the local gauge-degenerate alternative. □

Theorem 4.3 (Local multibubble and cascade exclusion)**.** *Assume the local Type I tangent criterion and the local multibubble/cascade decoupling and scale-rigidity theorems. Then no local multibubble, cascade, or gauge-degenerate Type II configuration occurs.*

Proof recalled from Paper III. If compact-cylinder bounds fail, or if active CKN mass splits, another active bounded-cylinder concentration core is selected. Comparable cores are combined or remain in the multibubble case. Strict same-point cascades are reduced to scale-rigidity or nonlinear no-splitting. Separated physical cores decouple in active local frames. The local decoupling and scale-rigidity theorems then reduce every such configuration either to the single-core criterion or to the Type I tangent criterion, both of which exclude it. □

Theorem 4.4 (Compact-cylinder Caccioppoli regularity)**.** *Let (V, P, a, b) solve the renormalized Navier–Stokes equation on compact cylinders, with local pressure reconstruction modulo functions of time, bounded critical local norm on the cylinders under consideration, and bounded or window-integrable modulation coefficients. Then the local Caccioppoli estimate gives uniform local windowed H^1 control. Consequently the rough-core alternative, meaning failure of this local windowed H^1 bound in a bounded renormalized core, cannot occur under these hypotheses.*

Proof recalled from Paper IV. The local energy inequality is pulled back through the absolutely continuous scale-translation chart. A compactly supported cutoff gives the local Caccioppoli estimate. The pressure term is estimated modulo constants by the compact-ball pressure decomposition and Calderon–Zygmund/harmonic bounds. The local L^2 and L^3 terms are controlled on bounded cylinders, and the modulation terms are controlled by the stated coefficient hypotheses. This yields the windowed H^1 bound. □

Theorem 4.5 (Scale-collapse exclusion alternatives)**.** *Let (V, P, a, b, λ) be a renormalized Type II rescaling satisfying*

$$\partial_\tau \log \lambda = a, \qquad \lambda(\tau) \to 0,$$

and suppose a localized L^2 core has a positive floor. Then

$$\int_{\tau_0}^{\infty} a_-(\tau) M_R(\tau)\, d\tau = \infty.$$

If the finite-cost class is controlled by this scale-drift cost, the finite-cost scale-collapse rescaling is excluded. Alternatively, on thick scale-collapse windows with compactness, modulation convergence, and a stationary omega-limit, the rescaling produces a nonzero stationary or generalized self-similar profile. In parameter ranges where the corresponding Liouville theorem excludes such profiles, the scale-collapse rescaling is excluded.

Proof recalled from Paper V. The identity $\partial_\tau \log \lambda = a$ implies

$$\int_{\tau_1}^{\tau_2} a(\tau)\, d\tau = \log \lambda(\tau_2) - \log \lambda(\tau_1).$$

If $\lambda(\tau) \to 0$, the right-hand side tends to $-\infty$. Hence $\int a_- = \infty$, and the positive localized L^2 floor gives divergence of $\int a_- M_R$. The cost conclusion follows from the finite-cost control hypothesis. The structural alternative follows by translating thick windows, using compactness and modulation convergence to pass to an autonomous reduced limit, and then applying the stationary omega-limit and Liouville hypotheses. □

5. Assembly theorem

Theorem 5.1 (Local Type II exclusion theorem)**.** *Let (u, p) be a suitable weak Navier–Stokes solution near (x_0, T). Assume the local Type I tangent criterion, the local multibubble/cascade decoupling and scale-rigidity theorems, the compact-cylinder Caccioppoli hypotheses, and the* *scale-collapse cost or Liouville hypotheses stated in Theorem 4.5. Then (x_0, T) is not a Type II singular point.*

Proof. Suppose, for contradiction, that (x_0, T) is a Type II singular point. By Theorem 2.1, there is a sequence of parabolic rescalings with positive local CKN concentration. Since the Type I tangent case is removed by Theorem 2.2, the sequence belongs to the local Type II regime.

Apply the compact-cylinder dichotomy, Theorem 3.1. If the compact-cylinder suitable package fails, then the sequence enters the local multibubble or cascade alternative, which is excluded by Theorem 4.3. Thus we may assume compact-cylinder suitable bounds.

If the localized scale-translation conditions are degenerate, the sequence is again in the multibubble or gauge-degenerate alternative and is excluded by Theorem 4.3. Otherwise Theorem 4.2 gives the renormalized scale-translation variables and the local pressure representation on the selected single core.

For this renormalized single-core sequence, consider the possible failures of the selected-window compact argument. If the localized Type II cost is finite, then Theorem 4.1 excludes the sequence. If local windowed H^1 control fails in a bounded renormalized core, Theorem 4.4 excludes that alternative under the compact-cylinder Caccioppoli hypotheses. If the renormalized scale collapses while a localized core remains nonzero, Theorem 4.5 excludes the scale-collapse alternative under the stated finite-cost or Liouville hypotheses.

These alternatives exhaust the positive local Type II concentration cases: compact single-core finite cost, multibubble/cascade or gauge degeneracy, rough-core loss of local windowed H^1, and scale collapse. Each case has been excluded, contradicting the assumed Type II singularity. □

Corollary 5.2 (Conditional exclusion after Type I)**.** *Assume the local Type I tangent criterion and all analytic hypotheses listed in Theorem 5.1. Then no singular point of a suitable weak solution can belong to the local Type II regime.*

Proof. This is Theorem 5.1 applied at each putative singular point after the positive local concentration reduction. □

6. Remaining analytic inputs

The assembled theorem leaves no hidden alternatives. Any failure of the conclusion must occur through failure of one of the stated PDE inputs:

- (i) the local Type I tangent criterion;
- (ii) compact-cylinder suitable compactness or the corresponding local multibubble/cascade exclusion;
- (iii) localized scale-translation nondegeneracy on a selected core;
- (iv) compact-cylinder Caccioppoli control and the pressure decomposition used in the rough-core theorem;
- (v) finite-cost control by the localized scale-drift cost, or the parameter-specific Liouville theorem for the stationary or generalized self-similar scale-collapse profile.

Thus the assembly theorem is a conditional Type II exclusion theorem in purely PDE terms: every local Type II concentration is either one of the listed alternatives or is excluded by the corresponding theorem.

References

[CKN82] Luis Caffarelli, Robert Kohn, and Louis Nirenberg. Partial regularity of suitable weak solutions of the navier–stokes equations. *Communications on Pure and Applied Mathematics*, 35(6):771–831, 1982.